

Linear Algebra

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1 Introduction

Use examples in all the sections!!

2 Recollections

2.1 Linear Independence

An indexed set of vectors $\{v_1, v_2, \dots, v_p\}$ in a vector space V is said to be **linearly independent** if the vector equation

$$c_1v_1 + c_2v_2 + \dots + c_pv_p = 0$$

has only the trivial solution:

$$c_1 = c_2 = \dots = c_p = 0.$$

For example, the standard basis of $\mathbb{R}^n$ is linearly independent.

2.2 Orthogonality

Two vectors u and v in $\mathbb{R}^n$ are **orthogonal** if their dot product is zero:

$$u \cdot v = 0.$$

This concept is fundamental in many areas, such as when performing the Gram-Schmidt process to obtain an orthogonal basis.

3 Vector Spaces

A **vector space** V over the real numbers R is a non-empty set of objects, called vectors, on which two operations are defined: vector addition and scalar multiplication. These operations must satisfy the following ten rules for all vectors $u, v, w \in V$ and scalars $c, d \in \mathbb{R}^n$:

1. **Closure under addition:** $u + v \in V$.
2. **Commutativity:** $u + v = v + u$.
3. **Associativity:** $(u + v) + w = u + (v + w)$.
4. **Existence of the zero vector:** There is a $0 \in V$ such that $u + 0 = u$.
5. **Existence of additive inverses:** For each $u \in V$, there exists $-u \in V$ with $u + (-u) = 0$.
6. **Closure under scalar multiplication:** $cu \in V$.
7. **Distributivity over vector addition:** $c(u + v) = cu + cv$.
8. **Distributivity over scalar addition:** $(c + d)u = cu + du$.
9. **Associativity of scalar multiplication:** $c(du) = (cd)u$.

10. **Identity element of scalar multiplication:** $1u = u$.

Thm. 1

If $v_1, \dots, v_p$ are vectors in a vector space V , then

$$\text{Span}\{v_1, \dots, v_p\}$$

is a subspace of V .

3.1 Column Spaces

The **column space** of an $m \times n$ matrix A , written as $\text{Col } A$, is the set of all linear combinations of the columns of A . If

$$A = [a_1, a_2, \dots, a_n],$$

then

$$\text{Col } A = \text{Span}\{a_1, a_2, \dots, a_n\}.$$

Thm. 2

The column space of an $m \times n$ matrix A is a subspace of R^m .

3.2 Null Spaces

The **null space** of an $m \times n$ matrix A , denoted as $\text{Nul } A$, is the set of all solutions to the homogeneous equation

$$A\mathbf{x} = 0.$$

In set notation,

$$\text{Nul } A = \{\mathbf{x} \in R^n : A\mathbf{x} = 0\}.$$

Thm. 3

The null space of an $m \times n$ matrix A is a subspace of R^n .

4 Spanning Sets

5 Bases

6 Gram-Schmidt Process

References